

Comparing Connectivity-Aware Losses for Retinal Vessel Segmentation

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Abstract

Retinal vessel segmentation is challenging because thin branches can be easily disconnected even when overall pixel-level overlap remains acceptable. In this project, we compared five loss settings under the same U-Net segmentation framework to study how loss design affects the models' ability to capture connecting structures. We defined a baseline model that provides robustness under general machine learning settings for class imbalanced data like retinal vessel data, and evaluated the segmentation performance on models trained with several combinations of loss functions, including Binary Cross Entropy, Dice, clDice, Skeleton Recall, and cbDice-like loss to draw insights into their segmentation behaviors through both evaluation metrics and visualization. Our results suggested that adding connectivity-aware loss terms upon a trusted overlap loss baseline during training effectively improved models' connectivity preservation especially for thin tubular patterns that are less emphasized to overlap-based segmentation loss function.

1 Introduction

Image segmentation [1] as a computer vision technique is widely used in a variety of

important and cutting-edge tasks like medical imaging [2], autonomous driving [3], remote sensing [4]. However, standard segmentation losses that rely mainly on pixel-wise or overlap-based objectives might not preserve the topology of images with thin, tubular, and highly connected structures like vessels, pulmonary airways, and roads on satellite images [3, 5, 6, 7, 8, 9, 10, 11]. Previous studies have shown that connectivity can be better preserved by introducing connectivity-aware loss functions [6, 5, 12, 11, 9, 13]. In this project, we investigate the effect of different loss functions on the ability of the trained models to correctly identify connectivity patterns in this type of image. Specifically, we conduct a comparison between the performance of 5 models with customized loss functions trained on retinal vessel datasets [7], to answer the question: Which loss function best preserves vessel connectivity under the same experimental setup?

Although pixel-wise classification accuracy is one way to evaluate model performance, its limitation in correctly identifying meaningful regions, such as a single pixel that connects two continuous segments, may still remain hidden. This is especially important in medical applications such as disease diagnosis [14], where small misclassified regions can disconnect segments and affect downstream analysis.

It is also interesting to consider whether applying connectivity-aware losses could positively affect segmentation, or even related generative tasks, in other images with similar characteristics, such as coronal loop images taken by the Solar Dynamics Observatory (SDO) [15].

Based on the project outcomes, we distinguish the segmentation behaviors of different loss functions under the same U-Net architecture [16], support them with numerical and visualization results, and draw insights into which loss may perform better under different scenarios. Through model comparison, we identify a loss combination that provides the most robust predictions in our experimental setting for images of retinal vessels from different data sources. We also custom-implemented several connectivity-aware evaluation metrics mentioned in previous published papers [17, 5, 9, 12, 13]. In a word, we conducted a horizontal comparison of connectivity-aware loss functions, with customized implementations, to establish a practical and easy-to-apply guideline for selecting effective loss functions in segmenting connected patterns.

2 Proposed Approach

2.1 Architecture

The overall project architecture followed a general machine learning architecture. We started by passing the raw data through a preprocessing pipeline so that the images and their corresponding ground-truth masks were converted into a unified format for training. The preprocessed retinal images were then fed into a fixed U-Net architecture (Figure. 1), while the vessel masks were used as labels for model learning under different loss functions. After training, we called the saved models to generate predicted vessel masks, which were further evaluated by two complementary criteria: quantitative evaluation metrics and qualitative visualization. The quantitative part was based on both standard segmentation metrics and connectivity-aware metrics, while the qualitative part compared the predicted masks

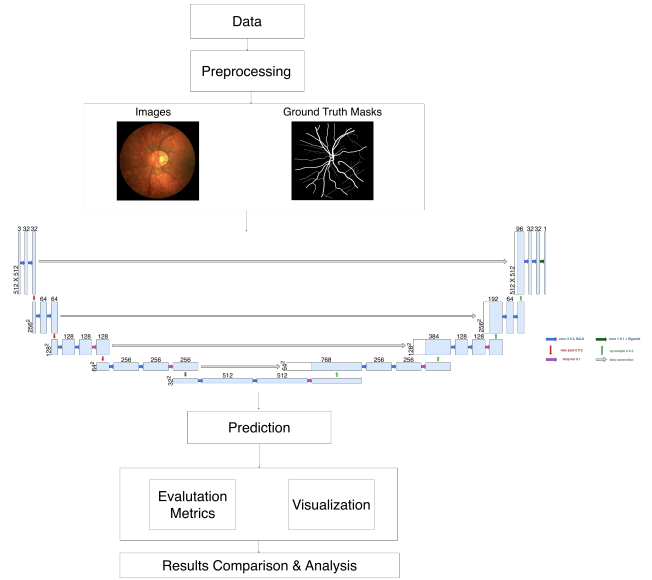


Figure 1: Overall architecture for this project, demonstrating the workflow from raw data, data preprocessing, U-Net-based segmentation, prediction, evaluation, to results analysis. The U-Net has an encoder that progressively downsamples the feature maps, while the decoder reconstructs the segmentation through up-sampling and skip connections.

with the ground-truth masks. Finally, we brought these results together in the section of results comparison and analysis, where we studied the behaviors of different loss functions under the same experimental setting to determine which one preserves vessel connectivity more effectively.

2.2 Design

The design of this project was based on a controlled comparison framework for studying the effect of loss functions on connectivity preservation in retinal vessel segmentation. As shown in Figure. 2, we combined the retinal vessel datasets from four data sources, including DRIVE [18], STARE [19], CHASE_DB1 [20], and HRF [21], and passed the paired images and masks through the same preprocessing procedure before splitting them into training, validation, and test sets. We fixed the architecture with the U-Net backbone and treated the loss function as the main experimental variable. Specifically, five loss functions were used to train separate models under the same general setup. We explicitly

defined the compared losses in mathematical form. BCE+Dice was used as the baseline loss, which combines a pixel-wise classification term with an overlap-based term. In our implementation, the Dice loss[17] is defined as

$$\mathcal{L}_{Dice} = 1 - \frac{2 \sum y\hat{y} + \epsilon}{\sum y + \sum \hat{y} + \epsilon},$$

and the BCE+Dice loss is written as

$$\mathcal{L}_{BCE+Dice} = \text{mean}(-y \log(\hat{y}) - (1 - y) \log(1 - \hat{y})) + \mathcal{L}_{Dice}$$

To further account for the class imbalance between vessel and background pixels, we also considered a weighted BCE term [22]. The corresponding loss used in this project is

$$\mathcal{L}_{WBCE+Dice} = \text{mean}(-(w y \log(\hat{y}) + (1 - y) \log(1 - \hat{y}))) + \mathcal{L}_{Dice}$$

To compare connectivity-aware training objectives, we introduced three additional losses on top of the same BCE+Dice baseline. For the clDice-based model, we defined a topology-aware term through a two-direction centerline agreement,

$$T_{prec} = \frac{\sum S(\hat{Y}) \cdot Y}{\sum S(\hat{Y}) + \epsilon}, \quad T_{sens} = \frac{\sum S(Y) \cdot \hat{Y}}{\sum S(Y) + \epsilon},$$

$$\mathcal{L}_{clDice} = 1 - \frac{2 T_{prec} T_{sens} + \epsilon}{T_{prec} + T_{sens} + \epsilon},$$

and the final compared loss is

$$\mathcal{L}_{BCE+Dice+clDice} = \mathcal{L}_{BCE+Dice} + 0.1 \mathcal{L}_{clDice}$$

For the Skeleton Recall setting, we followed the idea of encouraging the prediction to recover the ground-truth centerline [6]. The practical loss used in this project is

$$\mathcal{L}_{SkelRec} = 1 - \frac{\sum S(Y) \cdot \hat{Y}}{\sum S(Y) + \epsilon},$$

and the final compared loss is

$$\mathcal{L}_{BCE+Dice+SkelRec} = \mathcal{L}_{BCE+Dice} + 0.1 \mathcal{L}_{SkelRec}$$

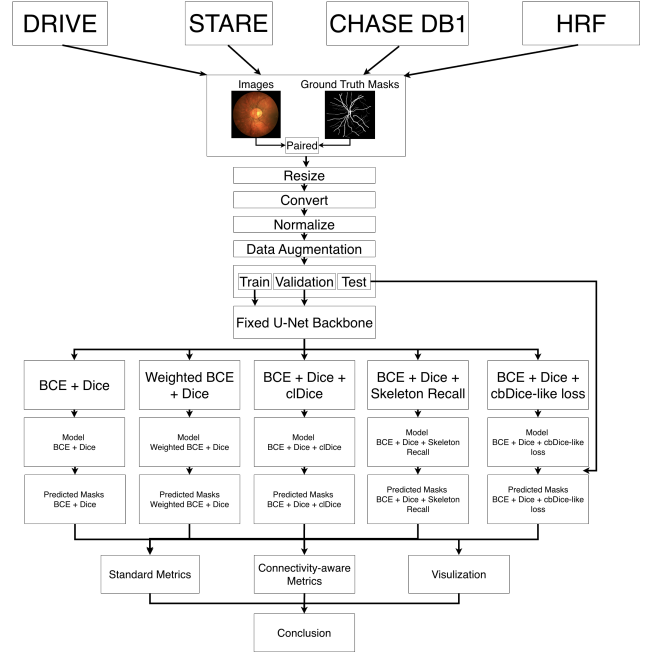


Figure 2: Design of the experimental framework in this project. Retinal images and ground-truth masks from DRIVE [18], STARE [19], CHASE_DB1 [20], and HRF [21] are paired, preprocessed, and split into training, validation, and test sets. Under the fixed U-Net backbone, five models trained with different losses are compared through their predicted masks. The final outputs are evaluated using standard metrics, connectivity-aware metrics, and visualization for final comparison.

For the cbDice-like setting, we combined a centerline term and a boundary term so that both vessel structure and vessel shape are taken into account [23]. The implemented centerline and boundary terms are

$$\text{CenterDice} = \frac{2 \sum S(Y) S(\hat{Y}) + \epsilon}{\sum S(Y) + \sum S(\hat{Y}) + \epsilon},$$

$$\text{BoundaryDice} = \frac{2 \sum B(Y) B(\hat{Y}) + \epsilon}{\sum B(Y) + \sum B(\hat{Y}) + \epsilon},$$

which gives the practical cbDice-like loss

$$\mathcal{L}_{cbDice} = 1 - [(1 - \beta) \text{CenterDice} + \beta \text{BoundaryDice}],$$

The final compared loss is

$$\mathcal{L}_{BCE+Dice+cbDice} = \mathcal{L}_{BCE+Dice} + 0.1 \mathcal{L}_{cbDice}$$

2.3 Implementation

The entire project code was developed in Python using Jupyter Notebook. We implemented the U-Net structure with Keras [24] and the compared loss functions with TensorFlow [25]. The input to the network is a $512 \times 512 \times 3$ retinal image, and the output is a $512 \times 512 \times 1$ vessel probability map [26] after sigmoid activation [27]. The model contains four encoder blocks with feature dimensions 32, 64, 128, and 256, followed by a bottleneck block with 512 channels. The decoder reconstructs the output through up-sampling, concatenation with skip connections, and convolutional blocks with feature dimensions 256, 128, 64, and 32 [16]. During training, we saved the best model checkpoint for each loss setting according to its validation performance. After training, we called the saved best models to generate predicted vessel masks on the held-out test set, then passed the outputs to the evaluation stage for numerical comparison and qualitative visualization.

3 Methodology

3.1 Data and Preprocessing

For data preparation, we paired each retinal image with its corresponding ground-truth mask and converted them into a common training format. All images were resized to 512×512 , normalized to the range $[0, 1]$, and matched with binary ground-truth masks of the same size, resulting in a total of 109 image-mask pairs: 20 pairs from DRIVE, 16 pairs from STARE, 28 pairs from CHASE_DB1, and 45 pairs from HRF. Since the combined dataset was collected from four different public sources, source-dependent variation still needed to be considered during shuffling and data splitting. In addition, because the cleaned dataset is relatively small and background boundaries appear differently across the four data sources, we applied light augmentations [28] to avoid the model taking background boundaries into account and

to prevent overfitting, while still preserving the detailed information carried by the data by not pushing the augmentation too hard. The augmentation techniques include random horizontal flipping and random vertical flipping with probability 0.5, small random rotation (around 10.8 degrees), mild brightness adjustment (with a maximum delta value of 0.08), and mild contrast adjustment (within the range from 0.9 to 1.1). These preprocessing and augmentation procedures were kept consistent across all compared models.

3.2 Baseline Model Selection

We first evaluated the baseline model through a 5-fold cross-validation [29] under BCE+Dice loss in order to examine whether its performance remained stable across different data splits. We observed that the validation behavior varied across folds, which suggests that the combined dataset still carries noticeable source-dependent variation and that model performance is sensitive to split composition. After this preliminary evaluation, we selected a fixed 73:12:15 train-validation-test split and kept it consistent throughout the remaining workflow. Under this setup, BCE+Dice was used as the baseline model, and the other four loss settings were compared against it.

3.3 Training Procedure

All models were trained under the same hyperparameter setting with batch size 4, learning rate 10^{-4} , and dropout [30] rate 0.1 in the deeper encoder, bottleneck, and early decoder layers. We set the maximum number of training epochs to 200, while using early stopping[31] to monitor the validation performance and save the best model checkpoint for each loss setting. As a common validation criterion across all compared losses, we used the validation soft Dice coefficient[32] to monitor performance and select the best checkpoint. This choice provides a shared overlap-based validation standard across different loss

settings and avoids directly comparing validation loss values defined on different loss scales. Since vessel segmentation is strongly affected by class imbalance between vessel and background pixels, soft Dice provides a more meaningful validation signal than accuracy for selecting the best checkpoint during training. Early stopping was applied with a tolerance of 15 epochs without improvement in validation soft Dice, starting from epoch 20, and the best weights were restored after training stopped. In addition, we applied automatic learning-rate reduction[33], where the learning rate was reduced by a factor of 0.5 if the validation soft Dice did not improve for 5 epochs, with a minimum learning rate of 10^{-6} .

4 Evaluation and Results

4.1 Validation Evaluation

Training curves for each loss setting were plotted across epochs on the validation set, as shown in Figure. 3. In general, the training and validation curves remain relatively close to each other for all five models, which suggests that the models were trained under a reasonably stable setting without obvious severe overfitting. Among them, the Weighted BCE+Dice model stopped earlier, around the 80th epoch, with validation soft Dice stabilizing near 0.60 and validation loss around 0.70. In contrast, the other four models continued training for a longer period, mostly beyond the 130th epoch, and reached stronger validation behavior, with validation soft Dice increasing to around 0.69–0.72 and validation loss decreasing to around 0.41–0.48. In particular, the BCE+Dice+clDice model showed the strongest validation performance among all settings, with validation soft Dice around 0.72 and validation loss around 0.41. Overall, these validation results suggest that all models reached a reasonable segmentation quality under the same training procedure, so the final test-stage comparison can be interpreted more meaningfully in terms of connectivity preservation rather than failure to achieve a basic

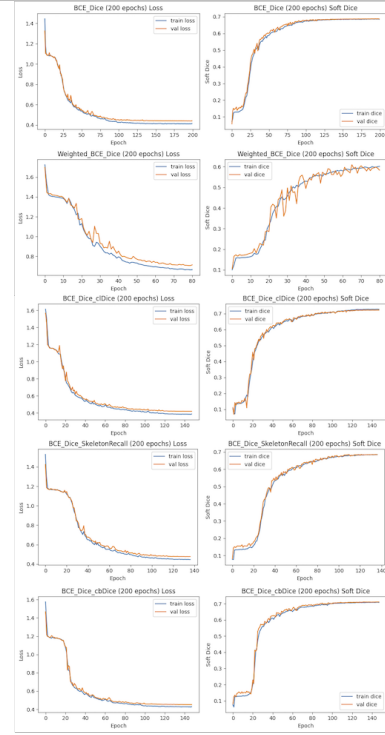


Figure 3: Training and validation curves for the five final loss settings. For each model, the left panel shows the training and validation loss across epochs, and the right panel shows the training and validation soft Dice coefficient across epochs. In general, the curves remain close between training and validation, which suggests that the compared models were trained under a relatively stable setting without obvious overfitting.

segmentation standard.

4.2 Evaluation Metrics

We evaluated the final models on the held-out test set using both standard segmentation metrics and connectivity-aware metrics, as summarized in Table 1. The standard metrics included Precision, Recall, Dice, and IoU[17], while the connectivity-aware metrics included clDice, connected-component error[12], skeleton recall[9], and Betti number error[13]. Since retinal vessel segmentation is strongly affected by class imbalance and thin connected structures, we emphasize metrics that reflect overlap and connectivity more directly rather than raw background agreement alone.

For the standard segmentation metrics,

Precision, Recall, Dice, and IoU are defined as

$$\text{Precision} = \frac{TP}{TP + FP + \epsilon},$$

$$\text{Recall} = \frac{TP}{TP + FN + \epsilon},$$

$$\text{Dice} = \frac{2TP}{2TP + FP + FN + \epsilon},$$

$$\text{IoU} = \frac{TP}{TP + FP + FN + \epsilon},$$

where TP , FP , and FN denote true positive, false positive, and false negative pixels, respectively. Intuitively, Precision measures how many predicted vessel pixels are truly vessels, Recall measures how much vessel structure is recovered, Dice measures the overlap between the prediction and the ground-truth mask, and IoU measures the overlap relative to their union.

For the connectivity-aware metrics, clDice is defined as

$$T_{prec} = \frac{|S(\hat{Y}) \cap Y|}{|S(\hat{Y})| + \epsilon}, \quad T_{sens} = \frac{|S(Y) \cap \hat{Y}|}{|S(Y)| + \epsilon},$$

$$\text{clDice} = \frac{2T_{prec}T_{sens}}{T_{prec} + T_{sens} + \epsilon},$$

where $S(\cdot)$ denotes the skeleton operator. In this metric, topology precision evaluates how much of the predicted centerline lies inside the true vessel region, while topology sensitivity evaluates how much of the true centerline is recovered by the prediction. Skeleton recall is defined as

$$\text{Skeleton Recall} = \frac{|S(Y) \cap \hat{Y}|}{|S(Y)| + \epsilon},$$

which measures how much of the ground-truth centerline is covered by the predicted mask. Connected-component error is defined as

$$\text{CC Error} = |C(\hat{Y}) - C(Y)|,$$

where $C(\cdot)$ denotes the number of connected foreground components. This metric reflects whether the prediction breaks vessel structures into too many disconnected segments or merges them incorrectly. In addition, we evaluated topological consistency through Betti number errors:

$$\beta_0 \text{ error} = |\beta_0(\hat{Y}) - \beta_0(Y)|,$$

$$\beta_1 \text{ error} = |\beta_1(\hat{Y}) - \beta_1(Y)|,$$

$$\text{Betti error} = \beta_0 \text{ error} + \beta_1 \text{ error},$$

where β_0 measures the number of connected components and β_1 measures the number of holes. Lower values of connected-component error and Betti error therefore indicate better preservation of global topology.

Based on the quantitative results in Table 1, BCE+Dice+clDice showed the strongest overall behavior under the current test setting. It obtained the highest Precision, Dice, IoU, clDice, and Skeleton Recall, together with the lowest β_1 error and Betti error, and also achieved the second-lowest connected-component error and β_0 error. BCE+Dice+cbDice showed the second-strongest overall behavior, with the second-highest Precision, Dice, IoU, clDice, and Skeleton Recall, while also obtaining the second-lowest β_1 error and Betti error. In contrast, Weighted BCE+Dice reached the highest Recall and the lowest connected-component error and β_0 error, but its lower Precision, Dice, and IoU suggest that this setting tends to recover more vessel pixels at the cost of introducing more false positives. Under the current experimental setup, these numerical results suggest that the clDice-based and cbDice-based losses are the most effective settings among the compared models for preserving vessel connectivity, while Weighted BCE+Dice reflects a stronger recall-oriented but noisier segmentation behavior.

Table 1: Final comparison table on the held-out test set. Bold values indicate the best performance in each metric, and underlined values indicate the second-best performance.

Model	Precision $\uparrow$	Recall $\uparrow$	Dice $\uparrow$	IoU $\uparrow$	clDice $\uparrow$	CC error $\downarrow$	SR $\uparrow$	β_0 error $\downarrow$	β_1 error $\downarrow$	Betti error $\downarrow$
BCE+Dice	0.716	0.733	0.722	0.567	0.733	827.625	0.643	827.625	573.250	1400.875
Weighted+BCE+Dice	0.602	0.766	0.670	0.507	0.725	795.750	0.673	795.750	577.188	1372.938
BCE+Dice+clDice	0.752	0.759	0.752	0.605	0.765	798.000	0.689	798.000	547.125	1345.125
BCE+Dice+SkeletonRecall	0.706	0.737	0.719	0.563	0.735	808.188	0.646	808.188	574.062	1382.250
BCE+Dice+cbDice	<u>0.717</u>	<u>0.762</u>	<u>0.736</u>	<u>0.584</u>	<u>0.754</u>	805.812	<u>0.680</u>	805.812	<u>563.438</u>	<u>1369.250</u>

4.3 Visualization

To complement the numerical comparison, we further visualized the predicted masks from all five models together with the corresponding ground-truth masks for five representative test samples, as shown in Figure. 4 By direct visual inspection, the models show distinguishable segmentation behaviors, especially on thin sub-branches and vessel tips where connectivity preservation becomes more difficult. In general, BCE+Dice is able to recover the major vessel skeletons correctly, but it misses a noticeable portion of thin vessel tips and small sub-branches. This behavior can be observed, for example, in the third row of Figure. 4, where the right-pointing thin branch is much weaker in the BCE+Dice prediction than in the ground truth.

Weighted BCE+Dice tends to recover more vessel pixels, but it also introduces more scattered false-positive responses. This can be seen from the noisy isolated spots that appear around the vessel structures in several rows, especially in the third row. This visual behavior is consistent with its quantitative results, where Recall is improved but Precision, Dice, and IoU are reduced. The BCE+Dice+Skeleton Recall model behaves similarly to BCE+Dice in its overall vessel coverage, but it also tends to produce more sparse non-connecting spots near fine vessel tips, which weakens the overall cleanliness of the segmentation.

Among all compared models, BCE+Dice+clDice produces the visually cleanest thin branches while still preserving their connectivity. It shows the strongest ability to keep fine sub-branches connected to the main vessel structures, especially in the lower-

left and outer-tip regions. BCE+Dice+cbDice shows a very similar behavior and also preserves thin branches well, but it is slightly weaker than BCE+Dice+clDice in overall correctness on several fine sub-branches. This difference can be observed in the fifth row, where the thin lower-left vessel tip is better preserved under the clDice-based setting. Overall, the visualization comparison supports the numerical results in Table 1 and shows that clDice- and cbDice-based losses provide the most consistent preservation of vessel connectivity under the current experimental setting.

4.4 Ablation Study

We carried out a set of ablation studies under the same experimental setting to examine how the main components in the baseline loss affected the segmentation behavior. Since BCE+Dice was used as the baseline model, the ablation study was designed to test whether either of its two major components could independently support a reasonable vessel segmentation under the current setup, and whether connectivity-aware terms alone could produce comparable results.

In the first ablation setting, we trained a model using BCE loss alone, the training process stopped around the 15th epoch. To further inspect its behavior, we also conducted a small overfitting test[34, 35] as a sanity check, by allowing the training process to continue up to 100 epochs without relying on early stopping. The purpose of this test was to check whether the model could at least fit the data more closely under a relaxed stopping condition. However, even under this setting, the validation soft Dice coefficient remained around

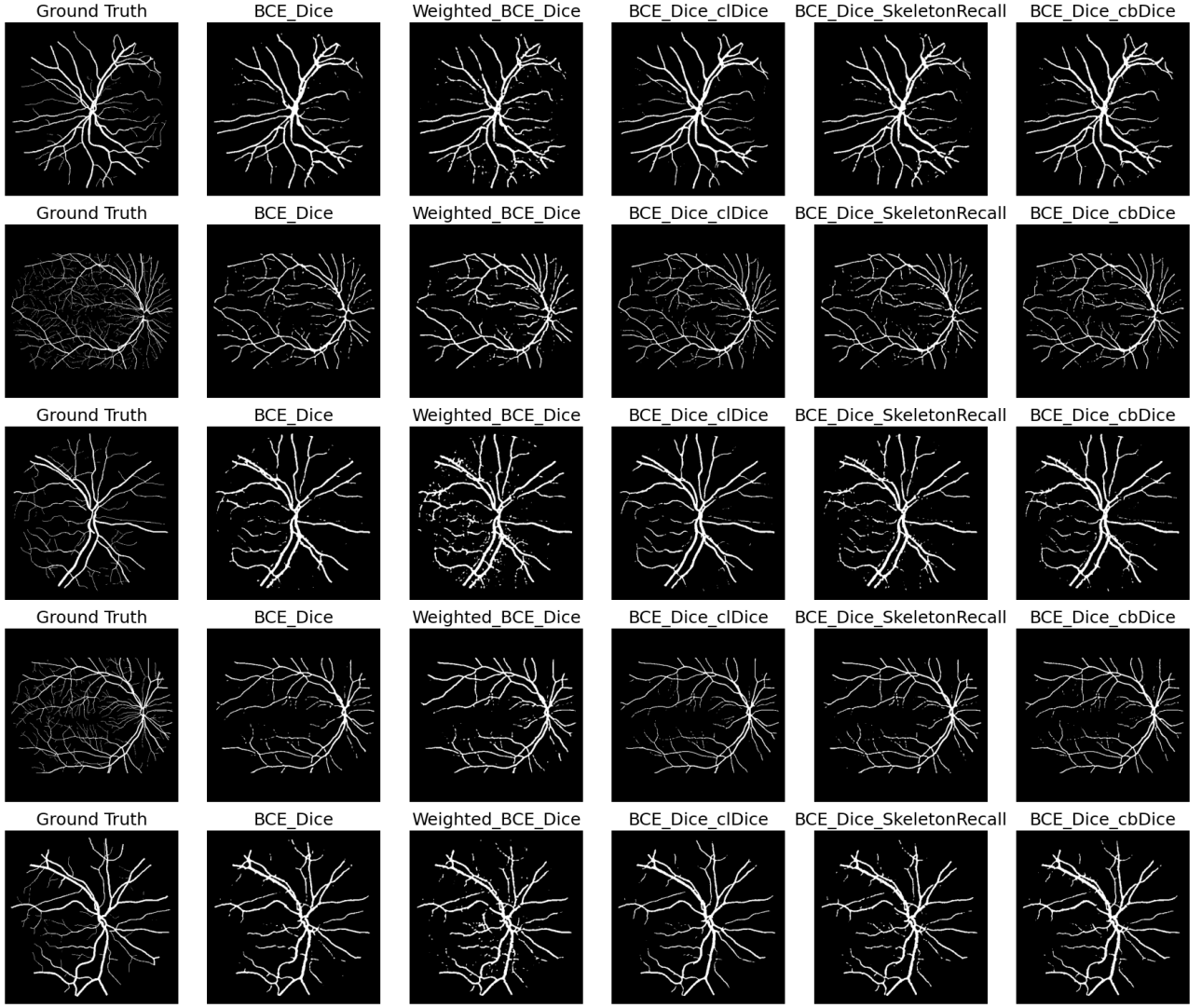


Figure 4: Visualization comparison of predicted vessel masks on five representative test samples. Each row corresponds to one test image, and the columns show the ground-truth mask and the predictions generated by the five final compared models. The comparison highlights differences in vessel continuity, thin-branch recovery, and false-positive patterns across loss settings.

0.35 in the later epochs, and the predicted masks indicated that the model tended to classify most of the retinal area as a large white disk rather than forming vessel-like structures. Under our current dataset and training setup, this suggests that BCE loss alone was not sufficient to recover the thin and sparse vessel patterns in a meaningful way.

A similar behavior was observed when Dice loss was used independently. In addition, when connectivity-aware terms such as clDice, Skeleton Recall, and cbDice-like components were added on top of Dice alone, the resulting models still did not produce vessel masks with a comparable level of structural correctness

to the final five selected models. Under the current experimental setup, these observations suggest that the connectivity-aware terms by themselves were not able to provide a stable segmentation behavior without a stronger baseline objective.

Therefore, within the scope of our current dataset and implementation setting, the BCE+Dice combination appears to serve as a more reliable baseline for the later connectivity-aware comparisons. The ablation results mainly show that the remaining comparisons in this study should be understood as refinements built on top of a baseline that already provides a reasonable segmentation be-

havior, rather than as independent replacements for it.

Conclusion

In this project, we conducted a controlled comparison of five loss settings for retinal vessel segmentation under the same general experimental setup, with the goal of examining how loss design affects connectivity preservation in thin and highly connected structures. By combining both standard segmentation metrics and connectivity-aware metrics, together with visualization comparison, we observed distinguishable segmentation behaviors across the compared models. Under the current dataset and implementation setting, BCE+Dice+clDice showed the strongest overall performance in terms of balancing segmentation correctness and connectivity preservation, while BCE+Dice+cbDice also showed competitive behavior. In contrast, Weighted BCE+Dice tended to recover more vessel pixels but also introduced more false-positive responses, which reflects a different tradeoff between recall and structural cleanliness.

From the ablation results, we also observed that BCE or Dice alone did not provide a stable enough segmentation behavior for the current dataset, and connectivity-aware terms alone were not sufficient to replace a stronger baseline objective under the same general hyperparameter setting. In this sense, the study suggests that the role of connectivity-aware losses in our setup is better understood as refining connectivity on top of a baseline that already provides a reasonable segmentation quality, rather than serving as a fully independent training objective.

At the same time, several limitations should be noted. First, the compared connectivity-aware losses were implemented as practical versions based on the main mathematical ideas from previous papers, but not as strict reproductions of their full original frameworks. In particular, our Skeleton Recall implementation follows the main intuition of recovering the ground-truth skeleton, but does

not include the complete framework proposed in the referenced paper, such as its focal variant and CPU-based computational design[6]. Similarly, our cbDice setting should be understood as a cbDice-like approximation, since the published version includes additional geometric and radius-aware mechanisms[23] that were not reproduced in our implementation. Therefore, the current findings should be interpreted as evidence under our practical implementation setting rather than as a complete benchmark of the original published methods.

Second, the weights for the added connectivity-aware loss terms were not extensively tuned, and the small coefficients used in this project were mainly selected as safe choices for maintaining stable training. Further studies may tune these weights more systematically in order to examine whether the observed ranking among the compared losses remains consistent across different weighting settings. Another limitation observed in the experiments is that model performance was not fully stable across all folds during the preliminary 5-fold cross-validation. Since the dataset was combined from four different public sources, differences in image style, background boundaries, and vessel appearance may have contributed to this split sensitivity. This suggests that the current conclusions should be interpreted within the present experimental setting and may still be affected by dataset heterogeneity.

For future work, it would be meaningful to reproduce the connectivity-aware losses more strictly according to their original papers, to test whether the same ranking still appears under a more faithful implementation. It would also be useful to extend the comparison to larger and more homogeneous retinal vessel datasets, or to additional segmentation backbones, in order to examine how stable these observations remain across different settings. More generally, the comparison framework developed in this project may also be useful for studying connected-pattern segmentation problems beyond retinal vessels, including images with similar thin and structured pat-

terns such as pulmonary airways, roads, and solar coronal loops.

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